Galaxy Rotation Curve Lab

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Abstract

In this lab, my lab partner and I observed a galaxy using a spectrograph to extract a rotation curve from it and compare it to models of rotation curves of a galaxy with and without dark matter. We compared how well each of these models fit to our observed rotation curve and found that the model for a galaxy with a dark matter halo fit our data the best (with a lower chi-square value of 1.259 vs. 1.670 for the model with no dark matter), leading us to believe that the galaxy we observed does have a dark matter halo.

1 Introduction

For this lab, my lab partner and I attempted to observe a galaxy in order to create a two-dimensional spectrum of it, which we could use to extract a rotation curve by observing how a known wavelength spectral line varies in observed wavelength across the disk of the galaxy. With a rotation curve, we could then fit models to it for mass-distributions with and without dark matter to see which fits best, therefore providing evidence for or against the existence of dark matter in galaxies.

2 Observations

2.1 Instrument

The telescope used to take this data was the Shane three-meter telescope at Lick Observatory and its KAST Spectrograph Camera, which uses two ccd sensors to capture images, one in the blue wavelength domain (3300-5520 Angstroms) and another in the red wavelength domain (3800-10000 Angstroms). For this lab, we used the red side with a grating of 600/7500 (grooves/blaze).

2.2 Targets

The science target we used for this lab was UGC 9780, a galaxy with right ascension 15:14:01.0157 and declination +44 35 25.731.

The calibration target we used was a He Hg Cd Arc lamp.

2.3 Summary of Observations

The science target was exposed 5 times for 600 seconds, while the arc lamp was exposed 4 times. We also captured 15 bias exposures for 0 seconds each and 15 flatfields for 5 seconds each.

3 Data Processing and Extraction

3.1 Bias, Flat, and Units Corrections

We started processing our data by median combining all 5 of our galaxy exposures , all 15 bias exposures, all 15 flats, and all 4 arc lamp exposures to get one median combined image of our galaxy (UGC9780), one of the arc lamp, one of the bias frames, and one of the flatfields. These median combined images are shown below (see Figures 1, 2, 3, 4).

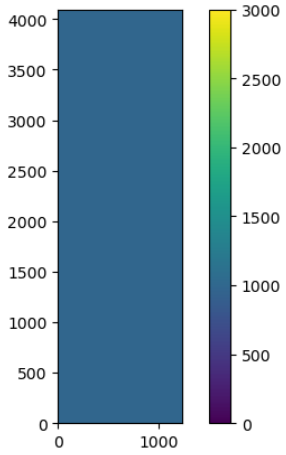


Figure 1: Median Combined Bias Image

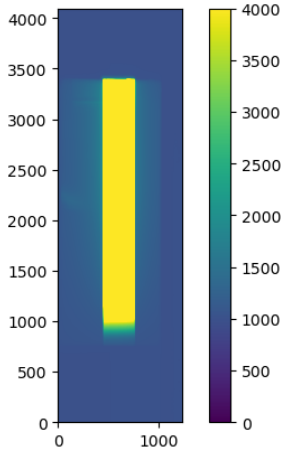


Figure 2: Median Combined Flatfield Image

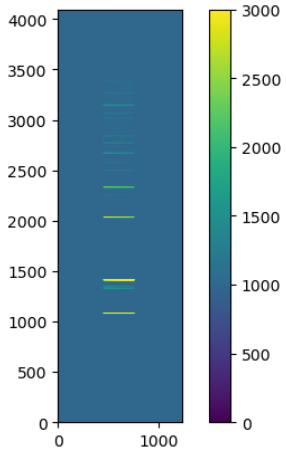


Figure 3: Median Combined Arc Lamp Image

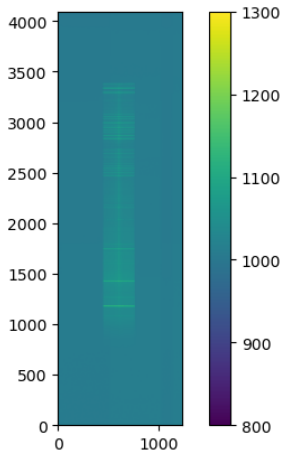


Figure 4: Median Combined Galaxy Image

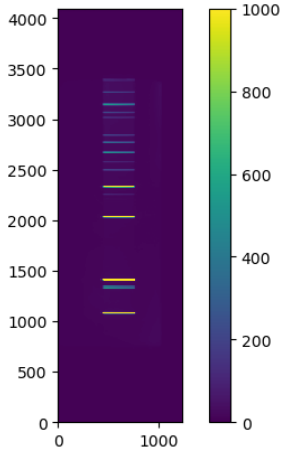


Figure 5: Cleaned Arc Lamp Spectrum

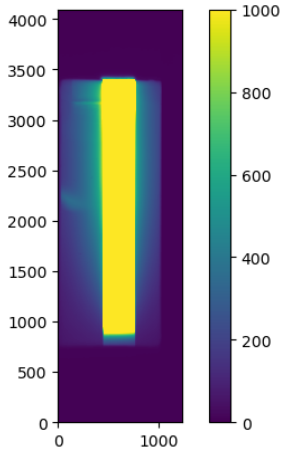


Figure 6: Cleaned Flat Field Spectrum

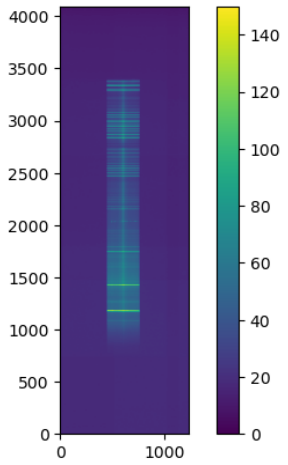


Figure 7: Cleaned Galaxy Spectrum

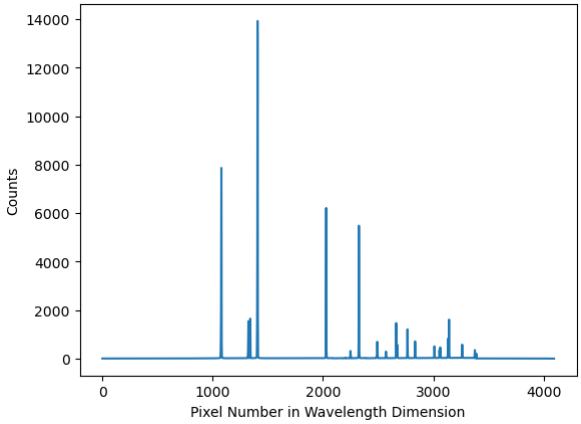


Figure 8: Average Counts vs. Pixel Number in the Wavelength Dimension of Arc Lamp

We then processed the images further by subtracting the median combined bias image from all the other median combined images, resulting in the images below (see Figures 5, 6, 7).

We then created a one-dimensional spectrum of the cleaned arc lamp frame by taking the median of each row only in the region illuminated by the arc lamp. This gave us a one-dimensional array only containing one count value per pixel in the wavelength dimension. Plotting this gave us a graph of average counts vs. pixel number in the wavelength dimension, as seen below (see Figure 8).

We then compared this arc spectrum to one in the Kast documentation with the same grating spacing, and we identified four spectral lines (the four tallest peaks in our spectrum) that we could compare to the known wavelengths in the Kast documentation spectrum. Plotting the known wavelengths of these lines vs. their pixel numbers, we found that there is a linear relationship between them, so we did a linear polynomial fit on these numbers, obtaining our wavelength calibration relation. Below is a plot of the pixel locations of these four spectral lines vs. their identified wavelengths, with the linear fit overlayed (see Figure 9).

Then, we applied the gain correction (1.9 e^-/DN) to the count numbers of the spectrum, changing the data numbers from counts to electrons. Applying the wavelength calibration relation and the gain correction to the arc spectrum resulted in the plot above (see Figure 10).

Then, we did the same process that we did for our cleaned arc lamp frame to our cleaned galaxy frame to create a one-dimensional spectrum. We then wavelength calibrated and gain corrected this galaxy spectrum and looked for the H-alpha emission line, which is at 6563 Angstroms at rest, and about 6676 Angstroms at our galaxy's expected redshift (0.0172) from the galaxy catalog. This was

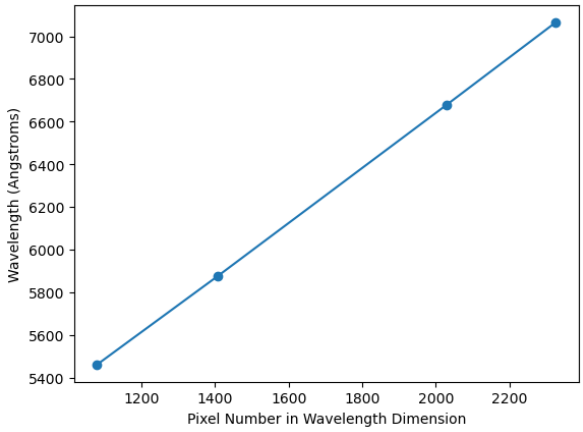


Figure 9: Identified Wavelength vs. Pixel Number in Wavelength Dimension

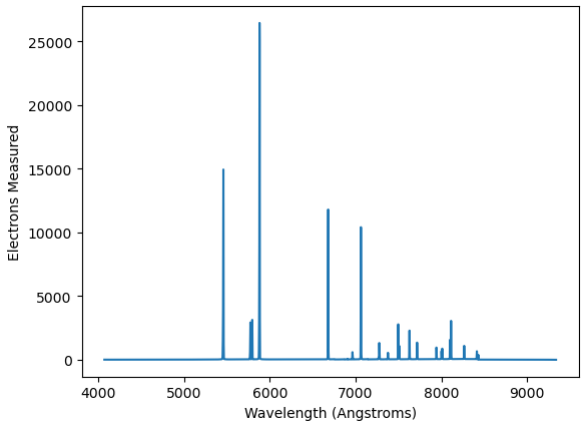


Figure 10: Electrons Measured vs. Wavelength (Angstroms)

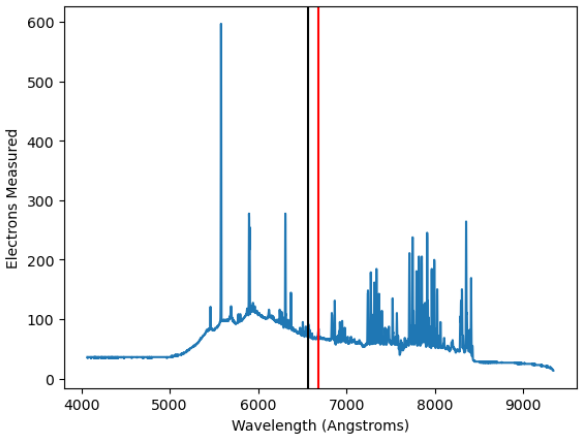


Figure 11: Electrons Measured vs. Wavelength (Angstroms) of UGC9780

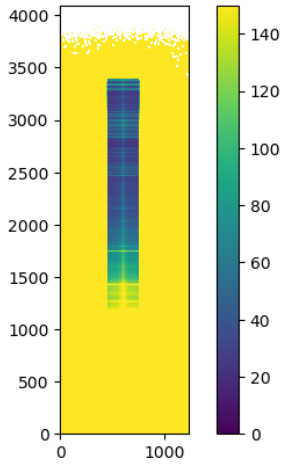


Figure 12: Flatfield Corrected Galaxy Spectrum in Counts

calculated using this expression (see Equation 1).

$$\lambda_{obs} = (z * \lambda_{rest}) + \lambda_{rest} \tag{1}$$

Where 'λ_{obs}' is the redshifted wavelength of the H-alpha line, 'z' is the galaxy's expected redshift, and 'λ_{rest}' is the rest wavelength of the H-alpha emission line.

The plot below shows the calibrated galaxy spectrum with a black vertical line representing the rest wavelength of the H-alpha line, and a red vertical line representing its red-shifted wavelength (see Figure 11).

With the H-alpha region identified, our next step was to apply the flatfield correction to the galaxy spectrum. First we normalize the flatfield in the region from 100 pixels below the H-alpha line to 100 pixels above it, since we only need to use the H-alpha line. Then we divided the galaxy spectrum image by the locally normalized flatfield, giving us a flatfield corrected galaxy spectrum image, as seen below (see Figure 12).

Then we repeated the same process that we used to create the one-dimensional spectrums of the clean galaxy image and the arc lamp on the flatfield corrected galaxy image, resulting in the image below with the same black and red lines representing the rest and red-shifted H-alpha wavelengths (see Figure 13).

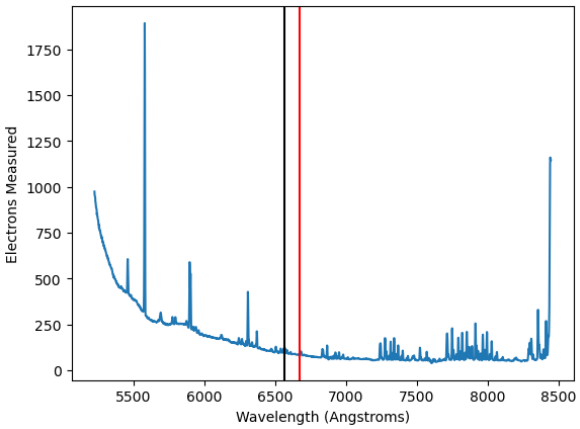


Figure 13: Electrons Measured vs. Wavelength (Angstroms) of UGC9780 (Calibrated)

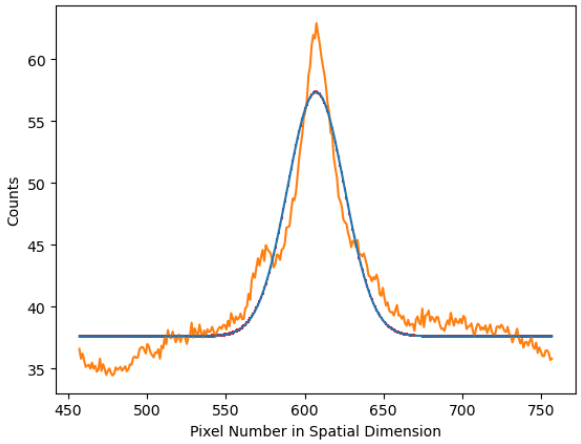


Figure 14: Counts vs. Pixel Number in Spatial Dimension

4 Analysis

With the one-dimensional spectrum of our galaxy made and our H-alpha emission line identified, our next goal was to extract a rotation curve. Our first step in doing this was to find the center of our galaxy by taking 100 rows in the wavelength dimension near the H-alpha line and median combining these to get a one-dimensional array with just the counts of the galaxy center in the middle. We then fit a Gaussian of counts vs. pixel in the spatial dimension to the one-dimensional array, centered approximately on the galaxy's center. One of the parameters given in the fit gave us the pixel value of the Gaussian's peak, which is approximately where the center of the galaxy is. The pixel number in the spatial dimension of the center of our galaxy ended up being 602.238 ± 0.171. Below is the plot of the one-dimensional array containing just the galaxy center (orange) with the Gaussian fit overlayed on top of it (blue) (see Figure 14).

Next, we identified the H-alpha line on the flatfield corrected galaxy exposure and cropped down the image to a twenty pixel wide region in the wavelength dimension containing it. This region ended up spanning pixels 2020 to 2040 in the wavelength dimension and 452 to 752 in the spatial dimension. This correlates to a wavelength range of 6668.656 ± 9.274 to 6694.401 ± 9.911 Angstroms. Below is a two-dimensional histogram of this region (see Figure 15).

Then, we began to extract the rotation curve out of this region by dividing the image into ten pixel wide bins in the spacial direction. For each bin, we then made a one-dimensional array containing the average counts of each ten pixel wide row and fit a Gaussian to it to determine the center of the H-alpha line in each bin. Then, we plotted the wavelength of the center of the H-alpha line in each bin vs. the pixel number of the center of each bin, which resulted in the following graph (see Figure 16).

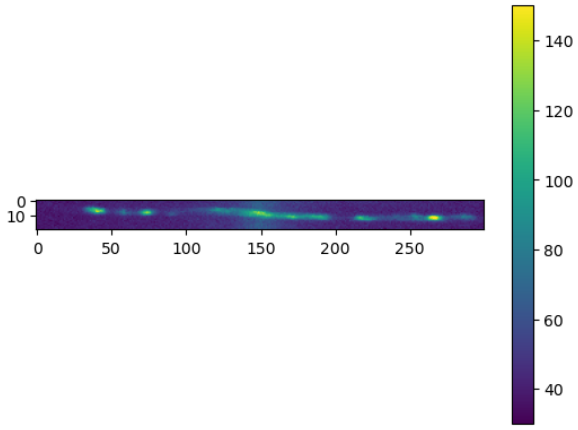


Figure 15: Region of Spectrum Containing the H-alpha Line

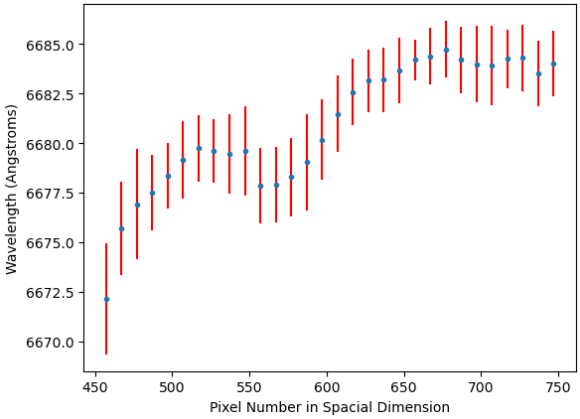


Figure 16: Wavelength of H-alpha Line vs. Pixel Number in Spacial Dimension

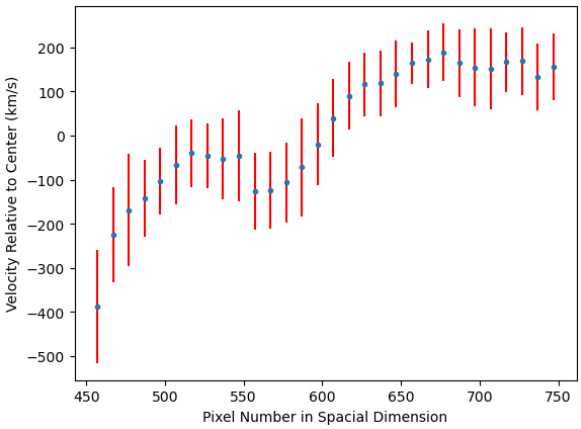


Figure 17: Velocity Relative to Galaxy Center vs. Pixel Number

The next step was to calculate the redshift of our galaxy, which we did by taking the wavelength of the H-alpha line in the central bin and using it in this equation (see Equation 2).

$$z = (\lambda_{obs} - \lambda_{rest})/\lambda_{rest} \tag{2}$$

Where 'z' is the redshift factor, 'λ_{obs}' is the wavelength of the H-alpha line in the central bin and 'λ_{rest}' is the rest wavelength of the H-alpha emission line.

Using this equation resulted in a redshift of z = $0.017921 \pm 9.251 * 10^{-8}$ (very close to the 0.0172 from the galaxy catalog), which when multiplied by the speed of light gives us the velocity the galaxy is traveling away from us, resulting in a velocity of 5372660 ± 27.733 m/s.

Next, we converted the wavelength values in the previous rotation curve to velocity relative to the center of the galaxy by applying the steps we used on the central bin to all bins and subtracting the velocity of the center of the galaxy from the velocity calculated for each bin. This gives us a new number for each bin that represents the velocity relative to the center of the galaxy of each bin's section of the galaxy. Plotting these relative velocities vs. the pixel number in the spacial dimension gives the following plot (see Figure 17).

To finalize our rotation curve, we then converted the pixel numbers in the spacial dimension of the center of each bin to distance from the center of the galaxy in kiloparsecs. We did this by first converting from pixels to arcseconds using the conversion rate given in the Kast red ccd documentation (0.43 arcseconds/pixel) and then using a public cosmology calculator and our measured redshift to calculate a conversion rate from arcseconds to kpc, which ended up being 0.366 kpc/arcsecond. Applying these conversion rates gave us an x-axis starting at zero and going up to the width of our galaxy in kpc, so to convert this axis to distance from the center, I just subtracted it by the distance from the left edge to the center. Finally, plotting the relative velocities vs. this new x-axis of distance from the galaxy's center resulted in this (see Figure 18).

With a complete rotation curve, I then "folded" the curve so that the x-axis only showed the absolute value of the distance from the center of the galaxy and the y-axis showed the absolute value of the relative velocity compared to the center. Then, I averaged the relative velocities of both sides to obtain an average rotation curve of our galaxy. The figures below show the folded curve with both sides of the galaxy and the averaged curve of the galaxy respectively (see Figures 19, 20).

The next step was to fit some rotation curve models to the average velocity curve, starting with a model for a mass distribution that is an exponential disk, not accounting for dark matter. The model uses this expression for circular velocity, or velocity relative to the center (see Equation 3).

$$v_d^2 = 4\pi G \Sigma_0 R_0 y^2 [I_0(y)K_0(y) - I_1(y)K_1(y)] \tag{3}$$

Where G is the gravitational constant, Σ_0 is a surface mass density and the parameter we are trying to fit the model for, R_0 is the scale length of the disk (maximum radius of the H-alpha line divided by four), y is the distance from the galaxy center, $I_i(y)$ is the modified Bessel function of the first kind of order i, and $K_i(y)$ is the modified Bessel function of the second kind of order i.

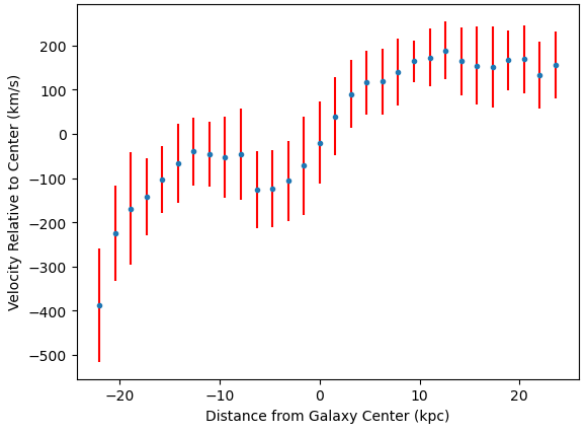


Figure 18: Velocity Relative to Galaxy Center vs. Distance from Galaxy Center

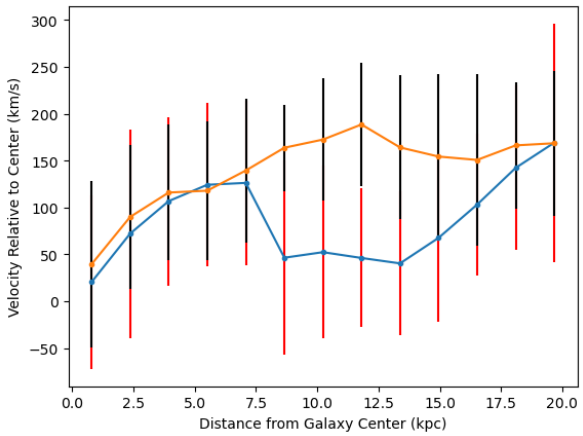


Figure 19: Velocity Relative to Galaxy Center for Left Side (Blue) and Right Side (Orange) vs. Absolute Distance from Galaxy Center

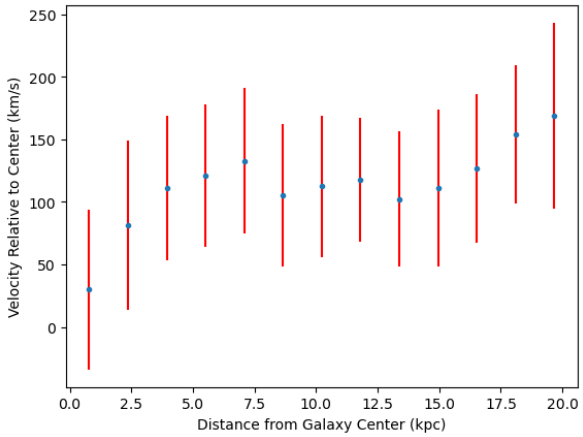


Figure 20: Average Velocity Relative to Center vs. Absolute Distance from Galaxy Center

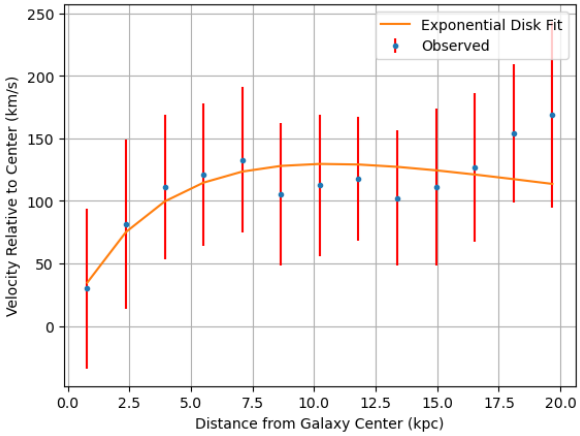


Figure 21: Average Velocity Relative to Center vs. Absolute Distance from Center with an Exponential Disk Model Overlayed

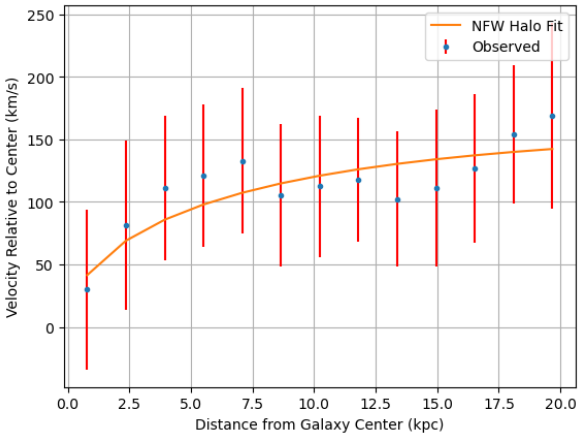


Figure 22: Average Velocity Relative to Center vs. Absolute Distance from Center with an NFW Halo Model Overlayed

Fitting this model to our data resulted in a Σ_0 of 326236110 ± 34235793.9 kg/kpc, producing a curve that looks like this (see Figure 21).

Next, we fit a model representing a spherically-symmetric dark matter halo that follows a Navarro–Frenk–White (NFW) distribution to the data. This model's expression for circular velocity is as follows (see Equation 4).

$$v_h^2 = GM(r)/r \tag{4}$$

Where r is the distance from the center and $M(r)$ is given by this equation (see Equation 5).

$$M(r) = 4\pi\rho_0 a^3[ln(1 + r/a) - (r/a)/(1 + r/a)] \tag{5}$$

With ρ_0 being a mass density and the parameter we are trying to fit for and a being a scale radius of 25 kpc.

Fitting this model resulted in a ρ_0 of 3363954.4 ± 314127.25 kg/kpc^3 and produced this curve (see Figure 22).

Then, I combined the two models to get a model of an exponential disk mass distribution plus the NFW dark matter halo and fit this composite model to the same data, now fitting for both Σ_0 and ρ_0. This resulted in a Σ_0 of 97987505 ± 130994320 kg/kpc and a ρ_0 of 2365034.3 ± 1357009.2 kg/kpc^3, producing this curve (see Figure 23).

Here is what our rotation curve looks like with all three models overlayed (see Figure 24).

To test for goodness of fit, we then did chi-square tests for all three models, resulting in a chi-square of $\chi^2_{Exp.Disk} = 1.670$ for the exponential disk model, $\chi^2_{NFW} = 1.331$ for the NFW dark matter halo

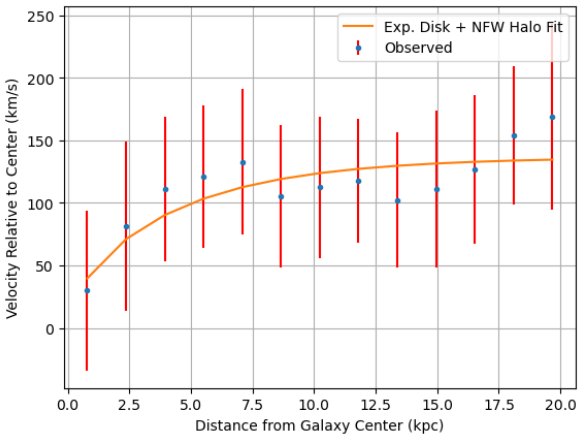


Figure 23: Average Velocity Relative to Center vs. Absolute Distance from Center with an Exponential Disk and NFW Halo Composite Model Overlayed

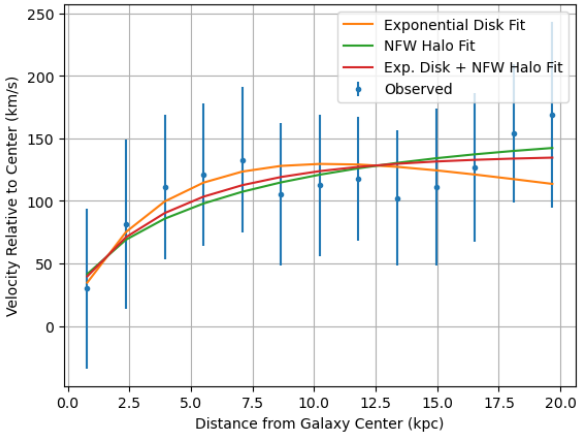


Figure 24: Average Velocity Relative to Center vs. Absolute Distance from Center with all Three Models Overlayed

model, and $\chi^2_{Comp.} = 1.259$ for the composite model. This suggests that the composite model is the most accurate, as its value of chi-square is the lowest.

We also calculated the total mass of the NFW dark matter halo using Equation 5 and of the disk from the exponential disk model alone using this equation (see Equation 6).

$$M_{disk,total} = 2\pi\Sigma_0 R_0^2 \tag{6}$$

This resulted in an NFW halo mass of $6.9136113 * 10^{11} \pm 3.9668914 * 10^{11}$ kg and a disk mass of $4.9580514 * 10^{10} \pm 1.7813112 * 10^{17}$ kg, meaning that the amount of mass in the dark matter halo alone is about an order of magnitude greater than the amount of mass in the disk alone.

4.1 Error Analysis

The noise in the flatfield corrected galaxy spectrum was used as the error in the pixel numbers in the wavelength dimension and was calculated per pixel using this equation (see Equation 7).

$$Noise = (counts + RN^2_{e^-}) \tag{7}$$

Where "counts" is the number of counts recorded for the pixel and RN_{e^-} is the read-noise per pixel of the sensor, which was 3.8 for us.

The errors on the pixel rotation curve were obtained in the Gaussian fitting process, being the fitted value for the standard deviation used in the Gaussian function. These were converted to wavelength using the wavelength conversion relation.

These errors were then converted to redshift by squaring them and multiplying by $1/\lambda^2_{rest}$, with λ_{rest} being the rest wavelength of the H-alpha emission line. Then, to convert from redshift to velocity, we just multiplied them by the speed of light.

The errors of the model fitting parameters Σ_0 and ρ_0 were given by the square root of the covariances for these parameters, obtained in the model fitting process.

The errors of the masses were obtained by plugging the error on Σ_0 into Equation 6 instead of Σ_0 for the disk and plugging the error on ρ_0 into Equation 5 instead of ρ_0 for the halo.

5 Conclusion

Overall, my results match relatively well with the expected results, with the composite model having the lowest chi-square value of the three (1.259 vs. 1.670 and 1.331 for the disk-only and halo-only models respectively). I also found that the mass of the dark matter halo alone was about an order of magnitude greater than the mass of the disk alone ($6.9136113 * 10^{11} \pm 3.9668914 * 10^{11}$ kg vs. $4.9580514 * 10^{10} \pm 1.7813112 * 10^{17}$ kg), which is similar to what I was expecting. The errors on the model fitting parameters Σ_0 and ρ_0, as well as on the masses, were much higher than expected however, and I suspect this comes from an error in what quantities I used to calculate these errors from the fitting process. The actual values of Σ_0, ρ_0, and the masses, however, seem quite accurate, as the models fit well.